

Test Points

Test point	Details
TP0	0V
TP1	12V
TP2	5V
TP3	Low when S7 is pressed

Circuit and working

As shown in Fig. 2, the quiz game controller circuit is built around the popular PIC16F877A microcontroller (IC1), driver ULN2003 (IC2), transistor BC547 (T1), 7-segment display (DIS1), some LEDs and switches.

PIC16F877A. This microcontroller,

which forms the heart of the circuit, has 8-bit processor with 8kB Flash memory, 368-byte data memory and 256-byte EEPROM. In addition, it has three timers. A 7-segment (common-anode) display is connected to the PIC controller through buffer ULN2003 on port C (RC0-RC6).

Six pushbutton switches are connected to port D (RD0-RD5) of the microcontroller. This port is used as the input port. The logic values of these switches are scanned by the program through this port at a very fast rate of about 6 microseconds. The microcontroller keeps scanning the positions of the switches and, when a switch shows logic 1, it sends a signal to port B (RB7) and the buzzer sounds.

Port B of the microcontroller acts as output port for the LEDs also. Six LEDs (LED1 through LED6) connected to port pins RB0 through RB5 of IC1 are used as visual indications. For example, if the third team presses the switch first, LED3 will glow along with sounding of the buzzer, and at the same time digit '3' will be displayed on DIS1.

Switch S7 is connected to pin 1 of microcontroller for manual reset. Resistors R1 and R2 and capacitor C4 form the power-on reset circuitry.

ULN2003. The 7-segment display DIS1 (SUNRA400CA or its equivalent) is driven by ULN2003 driver IC2. Pins RC0 through RC6 of microcontroller are connected to input pins IN1 through IN7 of IC2. Its output pins OUT1 through OUT7 are connected to the segments 'a' through 'g' of DIS1.

Power supply. The power supply is derived from 230V AC mains primary to 15V, 500mA secondary transformer X1. Bridge rectifier module BR1 rectifies secondary output of the transformer, capacitor C1 filters it and sends to input of 7812 regulator IC3. The 12V regulated DC output from IC3 is sent to input of 7805 regulator IC4 to power buzzer and DI51. The 5V DC from 7805 powers the microcontroller.

Software

The program for microcontroller is written in Basic and compiled using Proton IDE v2.0. After compilation of the code

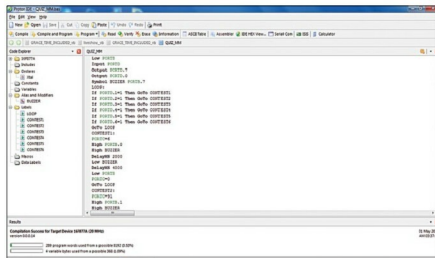


Fig. 3: Code compilation in Proton IDE

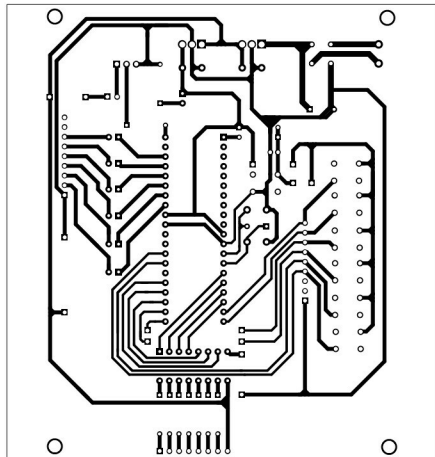


Fig. 4: Actual-size PCB layout for the quiz controller